Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Driving force

Meaning in my words How would you explain driving force to a friend?	**How can you tell?** A rope pulls right while friction acts left. Which one is the driving force on the cart?
Example or sketch Draw a 2 N driving pull and 0.4 N of friction. Which arrow is the driving force?	**Non-example and why** Is the hanging weight always the same as the rope tension on the cart?

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slope

Meaning in my words How would you explain slope to a friend?	**How can you tell?** Which cart has less mass if its acceleration rises more per newton?
Example or sketch Net force rises 4 N and acceleration rises 2 m/s². What mass fits that slope?	**Non-example and why** A partner reads the acceleration at one point and calls it slope. What two changes do you need?

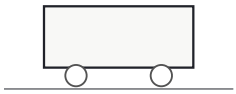
Friction offset

Meaning in my words How would you explain friction offset to a friend?	**How can you tell?** If the pull is 3 N and opposing friction is 1 N, how much force accelerates the cart?
Example or sketch For a sliding 2 kg cart with 1 N opposing friction, what acceleration would a 3 N pull produce?	**Non-example and why** A student uses the full 3 N pull as the net force. What did they leave out?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the pull and opposing friction. Label driving force on one arrow and net force on a separate arrow.



My marks mean

On assignment: Driving force

SECTION E / BACK / FINAL MEETING ONLY Name _____

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Flight-test engineer**

Draw two net-force arrows and two velocity-change arrows. Recommend the stronger change and state what you kept fixed.

☐ **Test-range broadcaster**

Tell viewers which run changes velocity more. Ask your producer whether doubling a motor's pull always doubles net force.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Distinguish a doubled applied force from a doubled net force.